Generate an MQL5 EA outline for ORB with order blocks

Here is a code-ready **MQL5 Expert Advisor (EA) outline** for an Open Range Breakout (ORB) strategy using SMC/ICT principles, specifically incorporating **order blocks** for entry logic. The structure below offers all function blocks, session management, order block marking, and key decision logic for you to expand into a full EA.

```
//+------------------------------------------------------------------+
//|           ORB + Order Block EA Outline (MQL5, 2025)              |
//+------------------------------------------------------------------+
input int       OR_start_hour = 8;        // Example: 8 (NY session)
input int       OR_start_minute = 30;     // 8:30 AM
input int       OR_range_minutes = 60;    // 1 hour range
input double    risk_per_trade = 1.0;     // % risk per trade

datetime OR_start, OR_end;
double OR_high, OR_low;

//--- For Order Block storage
struct OrderBlock {
    double price;
    datetime time;
    bool isBullish; // true=buy OB, false=sell OB
};
OrderBlock lastOB;

//+------------------------------------------------------------------+
//| Expert initialization function                                   |
//+------------------------------------------------------------------+
int OnInit() {
    OR_high = 0; OR_low = 0;
    OR_start = 0; OR_end = 0;
    lastOB.price = 0; lastOB.time = 0; lastOB.isBullish = false;
    return(INIT_SUCCEEDED);
}
//+------------------------------------------------------------------+
//| Main OnTick event                                                |
//+------------------------------------------------------------------+
void OnTick() {
    datetime now = TimeCurrent();

    // 1 - SESSION: Detect start of Opening Range
    if (TimeHour(now) == OR_start_hour && TimeMinute(now) == OR_start_minute && OR_start
        OR_start = now;
        OR_end = OR_start + OR_range_minutes * 60;
```

```
        OR_high = High[^0];
        OR_low = Low[^0];
    }

    // 2 - Update Opening Range High/Low
    if (OR_start > 0 && now >= OR_start && now <= OR_end) {
        if (High[^0] > OR_high) OR_high = High[^0];
        if (Low[^0] < OR_low) OR_low = Low[^0];
    }

    // 3 - After OR ends, search for Order Blocks at range boundaries
    if (now > OR_end) {
        // Find bullish/bearish OB near OR_high/low (simplified example:)
        if (Close[^1] > Open[^1] && Low[^1] == OR_low) { // Bullish OB at low
            lastOB.price = Low[^1];
            lastOB.time = Time[^1];
            lastOB.isBullish = true;
        } else if (Close[^1] < Open[^1] && High[^1] == OR_high) { // Bearish OB at high
            lastOB.price = High[^1];
            lastOB.time = Time[^1];
            lastOB.isBullish = false;
        }

        // 4 - ENTRY: Wait for breakout + retest of order block
        // Example long: price breaks OR_high, then retests bullish OB
        if (lastOB.isBullish && High[^0] > OR_high && Low[^0] <= lastOB.price && Close[^0
            // Calculate position size and send buy order
            // ...
        }

        // Example short: price breaks OR_low, then retests bearish OB
        if (!lastOB.isBullish && Low[^0] < OR_low && High[^0] >= lastOB.price && Close[^0
            // Calculate position size and send sell order
            // ...
        }
        // Reset for next day/session
    }
}

//--- Add functions for position sizing, order send, SL/TP, and risk controls as needed
//+------------------------------------------------------------------+
```

Key Features

- **Opening Range Session:** Detects and manages the range for the first hour (configurable).
- **Order Block Detection:** Marks possible OBs at OR high/low after the window closes (expansion suggested).
- **Breakout & Retest Logic:** Waits for price breakout, then entry on order block retest.
- **Skeleton for Order Management:** Ready to integrate custom SL/TP and position management.

Customization

- Refine *order block* logic using multi-candle structures or volume analysis for precision.
- Add full position sizing and risk control modules.
- Integrate multi-session support or push notifications for alerts.

This outline adopts standard SMC/ICT logic and is ready for you to expand for advanced automation or backtesting.[1] [2] [3]

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1. https://www.mql5.com/en/job/219692
2. https://www.mql5.com/en/job/219572
3. https://www.mql5.com/en/articles/16340